8

Techniques of Integration

8.1 Using Basic Integration Formulas

EXAMPLE 1 Evaluate the integral

$$\int_3^5 \frac{2x - 3}{\sqrt{x^2 - 3x + 1}}\,dx.$$

TABLE 8.1 Basic integration formulas

1. $\int k\,dx = kx + C$ (any number k)
2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
3. $\int \frac{dx}{x} = \ln|x| + C$
4. $\int e^x\,dx = e^x + C$
5. $\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0,\ a \neq 1)$
6. $\int \sin x\,dx = -\cos x + C$
7. $\int \cos x\,dx = \sin x + C$
8. $\int \sec^2 x\,dx = \tan x + C$
8. $\int \csc^2 x\,dx = -\cot x + C$
10. $\int \sec x \tan x\,dx = \sec x + C$
11. $\int \csc x \cot x\,dx = -\csc x + C$
12. $\int \tan x\,dx = \ln|\sec x| + C$
13. $\int \cot x\,dx = \ln|\sin x| + C$
14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$
15. $\int \csc x\,dx = -\ln|\csc x + \cot x| + C$
16. $\int \sinh x\,dx = \cosh x + C$
17. $\int \cosh x\,dx = \sinh x + C$
18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
19. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$
21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C \quad (a > 0)$
22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C \quad (x > a > 0)$

EXAMPLE 2 Complete the square to evaluate

$$\int \frac{dx}{\sqrt{8x - x^2}}.$$

EXAMPLE 3 Evaluate the integral

$$\int (\cos x \sin 2x + \sin x \cos 2x)\, dx.$$

EXAMPLE 5 Evaluate

$$\int \frac{3x^2 - 7x}{3x + 2}\,dx.$$

EXAMPLE 6 Evaluate

$$\int \frac{3x + 2}{\sqrt{1 - x^2}}\,dx.$$

8.2 Integration by Parts

Integration by parts is a technique for simplifying integrals of the form

$$\int u(x)\,v'(x)\,dx.$$

It is useful when u can be differentiated repeatedly and v' can be integrated repeatedly without difficulty. The integrals

$$\int x\cos x\,dx \qquad \text{and} \qquad \int x^2e^x\,dx$$

are such integrals because $u(x) = x$ or $u(x) = x^2$ can be differentiated repeatedly to become zero, and $v'(x) = \cos x$ or $v'(x) = e^x$ can be integrated repeatedly without difficulty. Integration by parts also applies to integrals like

$$\int \ln x\,dx \qquad \text{and} \qquad \int e^x\cos x\,dx.$$

In the first case, the integrand $\ln x$ can be rewritten as $(\ln x)(1)$, and $u(x) = \ln x$ is easy to differentiate while $v'(x) = 1$ easily integrates to x. In the second case, each part of the integrand appears again after repeated differentiation or integration.

Product Rule in Integral Form

If u and v are differentiable functions of x, the Product Rule says that

$$\frac{d}{dx}\left[u(x)\,v(x)\right] = u'(x)\,v(x) + u(x)\,v'(x).$$

In terms of indefinite integrals, this equation becomes

$$\int \frac{d}{dx}\left[u(x)\,v(x)\right]dx = \int \left[u'(x)\,v(x) + u(x)\,v'(x)\right]dx$$

or

$$\int \frac{d}{dx}\left[u(x)\,v(x)\right]\,dx = \int u'(x)\,v(x)\,dx + \int u(x)\,v'(x)\,dx.$$

Rearranging the terms of this last equation, we get

$$\int u(x)\,v'(x)\,dx = \int \frac{d}{dx}\left[u(x)\,v(x)\right]\,dx - \int v(x)\,u'(x)\,dx,$$

leading to the **integration by parts** formula

Integration by Parts Formula

$$\int u(x)\,v'(x)\,dx = u(x)\,v(x) - \int v(x)\,u'(x)\,dx \qquad (1)$$

Öğren

Integration by Parts Formula—Differential Version

$$\int u\,dv = uv - \int v\,du \qquad (2)$$

EXAMPLE 1 Find

$$\int x \cos x \, dx.$$

EXAMPLE 2 Find $\int \ln x \, dx$.

EXAMPLE 3 Evaluate

$$\int x^2 e^x \, dx.$$

EXAMPLE 4 Evaluate

$$\int e^x \cos x \, dx.$$

EXAMPLE 1 Find $\int x \sin x \, dx$.

V EXAMPLE 2 Evaluate $\int \ln x \, dx$.

EXAMPLE 5 Calculate $\int_0^1 \tan^{-1}x\,dx$.

EXAMPLE 6 Find the area of the region bounded by the curve $y = xe^{-x}$ and the x-axis from $x = 0$ to $x = 4$.

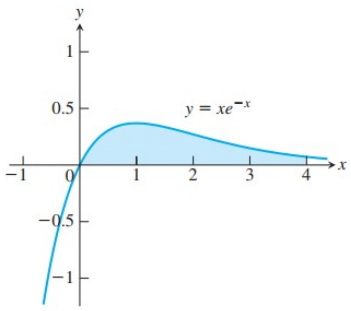


FIGURE 8.1 The region in Example 6.

EXAMPLE 6 Prove the reduction formula

$$\int \sin^n x\,dx = -\frac{1}{n}\cos x \sin^{n-1}x + \frac{n-1}{n}\int \sin^{n-2}x\,dx \quad [7]$$

where $n \geqslant 2$ is an integer.

EXAMPLE 5 Obtain a formula that expresses the integral

$$\int \cos^n x\,dx$$

in terms of an integral of a lower power of $\cos x$.

EXERCISES 8.2

Integration by Parts

Evaluate the integrals in Exercises 1–24 using integration by parts.

1. $\int x \sin \frac{x}{2}\, dx$
2. $\int \theta \cos \pi\theta\, d\theta$
3. $\int t^2 \cos t\, dt$
4. $\int x^2 \sin x\, dx$
5. $\int_1^2 x \ln x\, dx$
6. $\int_1^e x^3 \ln x\, dx$
7. $\int xe^x\, dx$
8. $\int xe^{3x}\, dx$
9. $\int x^2 e^{-x}\, dx$
10. $\int (x^2 - 2x + 1)e^{2x}\, dx$
11. $\int \tan^{-1} y\, dy$
12. $\int \sin^{-1} y\, dy$
13. $\int x \sec^2 x\, dx$
14. $\int 4x \sec^2 2x\, dx$
15. $\int x^3 e^x\, dx$
16. $\int p^4 e^{-p}\, dp$
17. $\int (x^2 - 5x)e^x\, dx$
18. $\int (r^2 + r + 1)e^r\, dr$

Evaluating Integrals

Evaluate the integrals in Exercises 31–56. Some integrals do not require integration by parts.

31. $\int x \sec x^2 \, dx$

32. $\int \frac{\cos \sqrt{x}}{\sqrt{x}} \, dx$

33. $\int x (\ln x)^2 \, dx$

34. $\int \frac{1}{x (\ln x)^2} \, dx$

35. $\int \frac{\ln x}{x^2} \, dx$

36. $\int \frac{(\ln x)^3}{x} \, dx$

37. $\int x^3 e^{x^4} \, dx$

38. $\int x^5 e^{x^3} \, dx$

39. $\int x^3 \sqrt{x^2 + 1} \, dx$

40. $\int x^2 \sin x^3 \, dx$

41. $\int \sin 3x \cos 2x \, dx$

42. $\int \sin 2x \cos 4x \, dx$

43. $\int \sqrt{x} \ln x \, dx$

44. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx$

45. $\int \cos \sqrt{x} \, dx$

46. $\int \sqrt{x} \, e^{\sqrt{x}} \, dx$

47. $\int_0^{\pi/2} \theta^2 \sin 2\theta \, d\theta$

48. $\int_0^{\pi/2} x^3 \cos 2x \, dx$

49. $\int_{2/\sqrt{3}}^{2} t \sec^{-1} t \, dt$

50. $\int_0^{1/\sqrt{2}} 2x \sin^{-1} (x^2) \, dx$

51. $\int x \tan^{-1} x \, dx$

52. $\int x^2 \tan^{-1} \frac{x}{2} \, dx$

58. Finding area Find the area of the region enclosed by the curve $y = x \cos x$ and the x-axis (see the accompanying figure) for

a. $\pi/2 \le x \le 3\pi/2$.

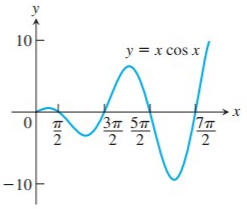
b. $3\pi/2 \le x \le 5\pi/2$.

c. $5\pi/2 \le x \le 7\pi/2$.

d. What pattern do you see? What is the area between the curve and the x-axis for

$$\left(\frac{2n-1}{2}\right)\pi \le x \le \left(\frac{2n+1}{2}\right)\pi,$$

n an arbitrary positive integer? Give reasons for your answer.



59. Finding volume Find the volume of the solid generated by revolving the region in the first quadrant bounded by the coordinate axes, the curve $y = e^x$, and the line $x = \ln 2$ about the line $x = \ln 2$.

60. Finding volume Find the volume of the solid generated by revolving the region in the first quadrant bounded by the coordinate axes, the curve $y = e^{-x}$, and the line $x = 1$

a. about the y-axis.

b. about the line $x = 1$.

8.3 Trigonometric Integrals

Products of Powers of Sines and Cosines

We begin with integrals of the form

$$\int \sin^m x \cos^n x \, dx,$$

where m and n are nonnegative integers (positive or zero). We can divide the appropriate substitution into three cases according to m and n being odd or even.

Case 1 If ***m* is odd**, we write m as $2k + 1$ and use the identity $\sin^2 x = 1 - \cos^2 x$ to obtain

$$\sin^m x = \sin^{2k+1} x = (\sin^2 x)^k \sin x = (1 - \cos^2 x)^k \sin x. \qquad (1)$$

Then we combine the single $\sin x$ with dx in the integral and set $\sin x \, dx$ equal to $-d(\cos x)$.

Case 2 If ***n* is odd** in $\int \sin^m x \cos^n x \, dx$, we write n as $2k + 1$ and use the identity $\cos^2 x = 1 - \sin^2 x$ to obtain

$$\cos^n x = \cos^{2k+1} x = (\cos^2 x)^k \cos x = (1 - \sin^2 x)^k \cos x.$$

We then combine the single $\cos x$ with dx and set $\cos x \, dx$ equal to $d(\sin x)$.

Case 3 If **both *m* and *n* are even** in $\int \sin^m x \cos^n x \, dx$, we substitute

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2} \qquad (2)$$

to reduce the integrand to one in lower powers of $\cos 2x$.

EXAMPLE 1 Evaluate

$$\int \sin^3 x \cos^2 x \, dx.$$

EXAMPLE 1 Evaluate $\int \cos^3 x \, dx$.

EXAMPLE 2 Find $\int \sin^5 x \cos^2 x \, dx$.

EXAMPLE 3 Evaluate $\int_0^{\pi} \sin^2 x \, dx$.

EXAMPLE 4 Find $\int \sin^4 x \, dx$.

Eliminating Square Roots

In the next example, we use the identity $\cos^2 \theta = (1 + \cos 2\theta)/2$ to eliminate a square root.

EXAMPLE 4 Evaluate

$$\int_0^{\pi/4} \sqrt{1 + \cos 4x}\, dx.$$

Integrals of Powers of tan x and sec x

We know how to integrate the tangent and secant functions and their squares. To integrate higher powers, we use the identities $\tan^2 x = \sec^2 x - 1$ and $\sec^2 x = \tan^2 x + 1$, and integrate by parts when necessary to reduce the higher powers to lower powers.

EXAMPLE 5 Evaluate

$$\int \tan^4 x \, dx.$$

EXAMPLE 5 Evaluate $\int \tan^6 x \sec^4 x \, dx$.

EXAMPLE 6 Find $\int \tan^5 \theta \sec^7 \theta \, d\theta$.

Products of Sines and Cosines

The integrals

$$\int \sin mx \sin nx\, dx, \qquad \int \sin mx \cos nx\, dx, \qquad \text{and} \qquad \int \cos mx \cos nx\, dx$$

arise in many applications involving periodic functions. We can evaluate these integrals through integration by parts, but two such integrations are required in each case. It is simpler to use the identities

$$\sin mx \sin nx = \frac{1}{2}[\cos(m-n)x - \cos(m+n)x], \tag{3}$$

$$\sin mx \cos nx = \frac{1}{2}[\sin(m-n)x + \sin(m+n)x], \tag{4}$$

$$\cos mx \cos nx = \frac{1}{2}[\cos(m-n)x + \cos(m+n)x]. \tag{5}$$

EXAMPLE 8 Evaluate

$$\int \sin 3x \cos 5x\, dx.$$

EXAMPLE 9 Evaluate $\int \sin 4x \cos 5x\, dx$.

7.2 EXERCISES

1–49 Evaluate the integral.

1. $\int \sin^3 x \cos^2 x \, dx$
2. $\int \sin^6 x \cos^3 x \, dx$
3. $\int_{\pi/2}^{3\pi/4} \sin^5 x \cos^3 x \, dx$
4. $\int_0^{\pi/2} \cos^5 x \, dx$
5. $\int \sin^2(\pi x) \cos^5(\pi x) \, dx$
6. $\int \frac{\sin^3(\sqrt{x})}{\sqrt{x}} \, dx$
7. $\int_0^{\pi/2} \cos^2\theta \, d\theta$
8. $\int_0^{\pi/2} \sin^2(2\theta) \, d\theta$
9. $\int_0^{\pi} \sin^4(3t) \, dt$
10. $\int_0^{\pi} \cos^6\theta \, d\theta$
11. $\int (1 + \cos\theta)^2 \, d\theta$
12. $\int x \cos^2 x \, dx$
13. $\int_0^{\pi/2} \sin^2 x \cos^2 x \, dx$
14. $\int_0^{\pi} \sin^2 t \cos^4 t \, dt$
15. $\int \frac{\cos^5\alpha}{\sqrt{\sin\alpha}} \, d\alpha$
16. $\int \cos\theta \cos^5(\sin\theta) \, d\theta$

Integrating Square Roots

Evaluate the integrals in Exercises 23–32.

23. $\displaystyle\int_0^{2\pi} \sqrt{\frac{1-\cos x}{2}}\,dx$

24. $\displaystyle\int_0^{\pi} \sqrt{1-\cos 2x}\,dx$

25. $\displaystyle\int_0^{\pi} \sqrt{1-\sin^2 t}\,dt$

26. $\displaystyle\int_0^{\pi} \sqrt{1-\cos^2\theta}\,d\theta$

27. $\displaystyle\int_{\pi/3}^{\pi/2} \frac{\sin^2 x}{\sqrt{1-\cos x}}\,dx$

28. $\displaystyle\int_0^{\pi/6} \sqrt{1+\sin x}\,dx$

$\left(\textit{Hint:}\ \text{Multiply by } \sqrt{\frac{1-\sin x}{1-\sin x}}.\right)$

29. $\displaystyle\int_{5\pi/6}^{\pi} \frac{\cos^4 x}{\sqrt{1-\sin x}}\,dx$

30. $\displaystyle\int_{\pi/2}^{3\pi/4} \sqrt{1-\sin 2x}\,dx$

31. $\displaystyle\int_0^{\pi/2} \theta\sqrt{1-\cos 2\theta}\,d\theta$

32. $\displaystyle\int_{-\pi}^{\pi} (1-\cos^2 t)^{3/2}\,dt$

Powers of Tangents and Secants

Evaluate the integrals in Exercises 33–50.

33. $\displaystyle\int \sec^2 x \tan x\,dx$

34. $\displaystyle\int \sec x \tan^2 x\,dx$

35. $\displaystyle\int \sec^3 x \tan x\,dx$

36. $\displaystyle\int \sec^3 x \tan^3 x\,dx$

37. $\displaystyle\int \sec^2 x \tan^2 x\,dx$

38. $\displaystyle\int \sec^4 x \tan^2 x\,dx$

39. $\displaystyle\int_{-\pi/3}^{0} 2\sec^3 x\,dx$

40. $\displaystyle\int e^x \sec^3 e^x\,dx$

8.4 Trigonometric Substitutions

Trigonometric substitutions occur when we replace the variable of integration by a trigonometric function. The most common substitutions are $x = a\tan\theta$, $x = a\sin\theta$, and $x = a\sec\theta$. These substitutions are effective in transforming integrals involving $\sqrt{a^2 + x^2}$, $\sqrt{a^2 - x^2}$, and $\sqrt{x^2 - a^2}$ into integrals we can evaluate directly since they come from the reference right triangles in Figure 8.2.

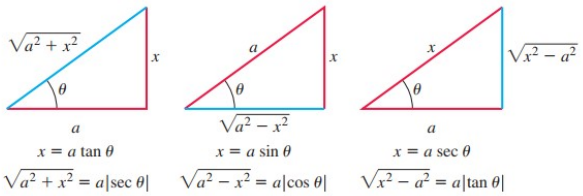


FIGURE 8.2 Reference triangles for the three basic substitutions identifying the sides labeled x and a for each substitution.

With $x = a\tan\theta$,

$$a^2 + x^2 = a^2 + a^2\tan^2\theta = a^2(1 + \tan^2\theta) = a^2\sec^2\theta.$$

With $x = a\sin\theta$,

$$a^2 - x^2 = a^2 - a^2\sin^2\theta = a^2(1 - \sin^2\theta) = a^2\cos^2\theta.$$

With $x = a\sec\theta$,

$$x^2 - a^2 = a^2\sec^2\theta - a^2 = a^2(\sec^2\theta - 1) = a^2\tan^2\theta.$$

$$x = a\tan\theta \quad \text{requires} \quad \theta = \tan^{-1}\left(\frac{x}{a}\right) \quad \text{with} \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2},$$

$$x = a\sin\theta \quad \text{requires} \quad \theta = \sin^{-1}\left(\frac{x}{a}\right) \quad \text{with} \quad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2},$$

$$x = a\sec\theta \quad \text{requires} \quad \theta = \sec^{-1}\left(\frac{x}{a}\right) \quad \text{with} \quad \begin{cases} 0 \le \theta < \dfrac{\pi}{2} & \text{if} \quad \dfrac{x}{a} \ge 1, \\ \dfrac{\pi}{2} < \theta \le \pi & \text{if} \quad \dfrac{x}{a} \le -1. \end{cases}$$

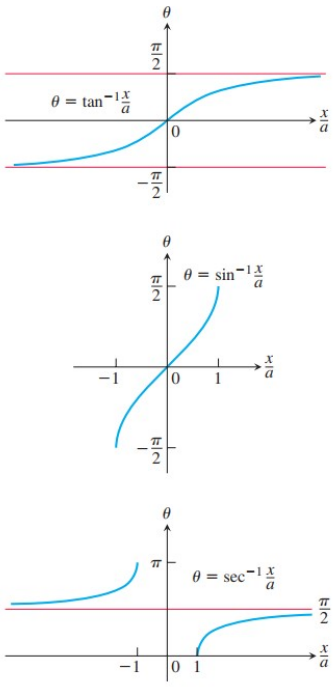


FIGURE 8.3 The arctangent, arcsine,

Procedure for a Trigonometric Substitution

1. Write down the substitution for x, calculate the differential dx, and specify the selected values of θ for the substitution.
2. Substitute the trigonometric expression and the calculated differential into the integrand, and then simplify the results algebraically.
3. Integrate the trigonometric integral, keeping in mind the restrictions on the angle θ for reversibility.
4. Draw an appropriate reference triangle to reverse the substitution in the integration result and convert it back to the original variable x.

EXAMPLE 1 Evaluate

$$\int \frac{dx}{\sqrt{4 + x^2}}.$$

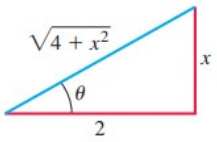


FIGURE 8.4 Reference triangle for $x = 2 \tan \theta$ (Example 1):

$$\tan \theta = \frac{x}{2}$$

EXAMPLE 3 Evaluate

$$\int \frac{x^2\,dx}{\sqrt{9-x^2}}.$$

EXAMPLE 4 Evaluate

$$\int \frac{dx}{\sqrt{25x^2-4}}, \qquad x > \frac{2}{5}.$$

EXAMPLE 4 Find $\int \dfrac{x}{\sqrt{x^2+4}}\,dx.$

EXAMPLE 6 Find $\int_0^{3\sqrt{3}/2} \dfrac{x^3}{(4x^2+9)^{3/2}}\,dx.$

EXAMPLE 7 Evaluate $\int \dfrac{x}{\sqrt{3-2x-x^2}}\,dx.$

7.3 EXERCISES

1–3 Evaluate the integral using the indicated trigonometric substitution. Sketch and label the associated right triangle.

1. $\int \frac{1}{x^2\sqrt{x^2-9}}\,dx; \quad x = 3\sec\theta$

2. $\int x^3\sqrt{9-x^2}\,dx; \quad x = 3\sin\theta$

3. $\int \frac{x^3}{\sqrt{x^2+9}}\,dx; \quad x = 3\tan\theta$

4–30 Evaluate the integral.

4. $\int_0^{2\sqrt{3}} \frac{x^3}{\sqrt{16-x^2}}\,dx$

5. $\int_{\sqrt{2}}^{2} \frac{1}{t^3\sqrt{t^2-1}}\,dt$

6. $\int_1^2 \frac{\sqrt{x^2-1}}{x}\,dx$

7. $\int \frac{1}{x^2\sqrt{25-x^2}}\,dx$

8. $\int \frac{x^3}{\sqrt{x^2+100}}\,dx$

9. $\int \frac{dx}{\sqrt{x^2+16}}$

10. $\int \frac{t^5}{\sqrt{t^2+2}}\,dt$

11. $\int \sqrt{1-4x^2}\,dx$

12. $\int_0^1 x\sqrt{x^2+4}\,dx$

21. $\int_0^{0.6} \frac{x^2}{\sqrt{9-25x^2}}\,dx$

22. $\int_0^1 \sqrt{x^2+1}\,dx$

23. $\int \sqrt{5+4x-x^2}\,dx$

24. $\int \frac{dt}{\sqrt{t^2-6t+13}}$

25. $\int \frac{x}{\sqrt{x^2+x+1}}\,dx$

26. $\int \frac{x^2}{(3+4x-4x^2)^{3/2}}\,dx$

27. $\int \sqrt{x^2+2x}\,dx$

28. $\int \frac{x^2+1}{(x^2-2x+2)^2}\,dx$

29. $\int x\sqrt{1-x^4}\,dx$

30. $\int_0^{\pi/2} \frac{\cos t}{\sqrt{1+\sin^2 t}}\,dt$

31. (a) Use trigonometric substitution to show that

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left(x+\sqrt{x^2+a^2}\right) + C$$

(b) Use the hyperbolic substitution $x = a\sinh t$ to show that

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C$$

These formulas are connected by Formula 3.11.3.

8.5 Integration of Rational Functions by Partial Fractions

$$\int \frac{5x - 3}{(x + 1)(x - 3)}\,dx = \int \frac{2}{x + 1}\,dx + \int \frac{3}{x - 3}\,dx$$

$$= 2\ln|x + 1| + 3\ln|x - 3| + C.$$

Paydayı çarpanlarına ayırma

The method for rewriting rational functions as a sum of simpler fractions is called **the method of partial fractions**. In the case of the preceding example, it consists of finding constants A and B such that

$$\frac{5x - 3}{x^2 - 2x - 3} = \frac{A}{x + 1} + \frac{B}{x - 3}. \qquad (1)$$

(Pretend for a moment that we do not know that $A = 2$ and $B = 3$ will work.) We call the fractions $A/(x + 1)$ and $B/(x - 3)$ **partial fractions** because their denominators are only part of the original denominator $x^2 - 2x - 3$. We call A and B **undetermined coefficients** until suitable values for them have been found.

General Description of the Method

Success in writing a rational function $f(x)/g(x)$ as a sum of partial fractions depends on two things:

- *The degree of $f(x)$ must be less than the degree of $g(x)$.* That is, the fraction must be proper. If it isn't, divide $f(x)$ by $g(x)$ and work with the remainder term. Example 3 of this section illustrates such a case.
- We must know the factors of $g(x)$. In theory, any polynomial with real coefficients can be written as a product of real linear factors and real quadratic factors. In practice, the factors may be hard to find.

Here is how we find the partial fractions of a proper fraction $f(x)/g(x)$ when the factors of g are known. A quadratic polynomial (or factor) is **irreducible** if it cannot be written as the product of two linear factors with real coefficients. That is, the polynomial has no real roots.

Method of Partial Fractions When $f(x)/g(x)$ Is Proper

1. Let $x - r$ be a linear factor of $g(x)$. Suppose that $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. Then, to this factor, assign the sum of the m partial fractions:

$$\frac{A_1}{(x - r)} + \frac{A_2}{(x - r)^2} + \cdots + \frac{A_m}{(x - r)^m}.$$

   Do this for each distinct linear factor of $g(x)$.

2. Let $x^2 + px + q$ be an irreducible quadratic factor of $g(x)$ so that $x^2 + px + q$ has no real roots. Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the n partial fractions:

$$\frac{B_1x + C_1}{(x^2 + px + q)} + \frac{B_2x + C_2}{(x^2 + px + q)^2} + \cdots + \frac{B_nx + C_n}{(x^2 + px + q)^n}.$$

   Do this for each distinct quadratic factor of $g(x)$.

3. Set the original fraction $f(x)/g(x)$ equal to the sum of all these partial fractions. Clear the resulting equation of fractions and arrange the terms in decreasing powers of x.

4. Equate the coefficients of corresponding powers of x and solve the resulting equations for the undetermined coefficients.

EXAMPLE 1 Find $\int \frac{x^3 + x}{x - 1}\,dx$.

$$\int \frac{x + 5}{x^2 + x - 2}\,dx$$

EXAMPLE 2 Evaluate $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x}\,dx$.

EXAMPLE 3 Find $\int \frac{dx}{x^2 - a^2}$, where $a \neq 0$.

EXAMPLE 2 Use partial fractions to evaluate

$$\int \frac{6x + 7}{(x + 2)^2}\,dx.$$

EXAMPLE 4 Use partial fractions to evaluate

$$\int \frac{-2x + 4}{(x^2 + 1)(x - 1)^2}\,dx.$$

EXERCISES 8.5

Expanding Quotients into Partial Fractions

Expand the quotients in Exercises 1–8 by partial fractions.

1. $\dfrac{5x - 13}{(x - 3)(x - 2)}$
2. $\dfrac{5x - 7}{x^2 - 3x + 2}$
3. $\dfrac{x + 4}{(x + 1)^2}$
4. $\dfrac{2x + 2}{x^2 - 2x + 1}$
5. $\dfrac{z + 1}{z^2(z - 1)}$
6. $\dfrac{z}{z^3 - z^2 - 6z}$
7. $\dfrac{t^2 + 8}{t^2 - 5t + 6}$
8. $\dfrac{t^4 + 9}{t^4 + 9t^2}$

Nonrepeated Linear Factors

In Exercises 9–16, express the integrand as a sum of partial fractions and evaluate the integrals.

9. $\displaystyle\int \frac{dx}{1 - x^2}$
10. $\displaystyle\int \frac{dx}{x^2 + 2x}$
11. $\displaystyle\int \frac{x + 4}{x^2 + 5x - 6}\,dx$
12. $\displaystyle\int \frac{2x + 1}{x^2 - 7x + 12}\,dx$
13. $\displaystyle\int_4^8 \frac{y\,dy}{y^2 - 2y - 3}$
14. $\displaystyle\int_{1/2}^1 \frac{y + 4}{y^2 + y}\,dy$
15. $\displaystyle\int \frac{dt}{t^3 + t^2 - 2t}$
16. $\displaystyle\int \frac{x + 3}{2x^3 - 8x}\,dx$

Repeated Linear Factors

In Exercises 17–20, express the integrand as a sum of partial fractions and evaluate the integrals.

17. $\displaystyle\int_0^1 \frac{x^3\,dx}{x^2 + 2x + 1}$
18. $\displaystyle\int_{-1}^0 \frac{x^3\,dx}{x^2 - 2x + 1}$
19. $\displaystyle\int \frac{dx}{(x^2 - 1)^2}$
20. $\displaystyle\int \frac{x^2\,dx}{(x - 1)(x^2 + 2x + 1)}$

Irreducible Quadratic Factors

In Exercises 21–32, express the integrand as a sum of partial fractions and evaluate the integrals.

21. $\displaystyle\int_0^1 \frac{dx}{(x + 1)(x^2 + 1)}$
22. $\displaystyle\int_1^{\sqrt{3}} \frac{3t^2 + t + 4}{t^3 + t}\,dt$
23. $\displaystyle\int \frac{y^2 + 2y + 1}{(y^2 + 1)^2}\,dy$
24. $\displaystyle\int \frac{8x^2 + 8x + 2}{(4x^2 + 1)^2}\,dx$
25. $\displaystyle\int \frac{2s + 2}{(s^2 + 1)(s - 1)^3}\,ds$
26. $\displaystyle\int \frac{s^4 + 81}{s(s^2 + 9)^2}\,ds$
27. $\displaystyle\int \frac{x^2 - x + 2}{x^3 - 1}\,dx$
28. $\displaystyle\int \frac{1}{x^4 + x}\,dx$
29. $\displaystyle\int \frac{x^2}{x^4 - 1}\,dx$
30. $\displaystyle\int \frac{x^2 + x}{x^4 - 3x^2 - 4}\,dx$
31. $\displaystyle\int \frac{2\theta^3 + 5\theta^2 + 8\theta + 4}{(\theta^2 + 2\theta + 2)^2}\,d\theta$
32. $\displaystyle\int \frac{\theta^4 - 4\theta^3 + 2\theta^2 - 3\theta + 1}{(\theta^2 + 1)^3}\,d\theta$

